

Safe Return to Sport Following Concussion



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KEYWORDS

• Concussion management • Concussion return to play • Return to sport

KEY POINTS

- New guidelines have been established to allow athletes to begin the stepwise progression for a return to sport with symptom-limited activity, such as walking, as early as 24 hours of injury.
- Better identification of concussions has allowed clinicians to initiate proper treatment plans and thus allow for decreased time loss and return to play.
- Concussions cannot be rushed, but new research can make health care professionals more comfortable by adding activity earlier than previously thought.
- Individualization is key in tailoring the steps to meet signs and symptoms and sport-specific needs.

INTRODUCTION

History and Definition of Concussion and Concussion Care

Concussions have been discussed in sports medicine to better understand how to diagnose and efficiently treat the injury, and how to return athletes to play most effectively. Concussion and mild traumatic brain injury can be used interchangeably if neuroimaging is considered normal or not indicated.¹ Concussion has historically been defined as “a physiological disruption of brain function with the possibility of microstructural brain injury” with other definitions being more detailed (**Box 1**).^{1–3} A neurometabolic cascade occurs after a concussion and refers to biochemical and physiological changes in the brain that cause an energy crisis, creating metabolic dysfunction.⁴ These initial changes have been thought to cause secondary injury and contribute to prolonged symptoms and recovery.⁴ Due to this, clinicians must focus their attention to effectively returning athletes to sport without causing further injury and prolonging time to return.

Collision sports have often been reported as having the highest incidence of concussion,⁵ with rugby as the highest concussion risk, followed by American football,

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Abbreviations	
ADLs	activities of daily living
ANS	autonomic nervous system
BP	blood pressure
CARE	Concussion Assessment, Research, and Education
CBF	cerebral blood flow
CCC	controlled chaos continuum
CISG	Concussion in Sport Group
HR	heart rate
NCAA	National Collegiate Athletic Association
RTL	return to learn
RTS	return to sport

ice hockey, and wrestling.⁶ Concussion rates had a steady increase from 2001 to 2018, with rates increasing by 0.19 per 10,000 athletic exposure each year.⁶ Concussion remains an ever-evolving injury that can occur in many populations in various instances. With large populations participating in sports and recreational activities, the risk of concussion and proper treatment of concussions remains a public health concern.⁶ Although concussion recognition has grown, there is limited awareness by the public regarding concussions and best practices, but increases in reporting, awareness, and diagnosis by physicians is promising.⁷

The first Concussion in Sport Group (CISG) convened in 2001 in Vienna as the first step to improving the health and safety of athletes regarding concussions and proposed a stepwise progression for return to play.⁸ Subsequent meetings of the CISG over the past 20 years have helped to identify varying components of concussion and have outlined 13 R's of concussion: recognition, removal, reevaluation, rest, rehabilitation, referral, recovery, return to sport (RTS), reconsiderations, residual effects and sequelae, and risk reduction,⁹ as well as the most recent additions of retirement and refinement.³

Signs and symptoms of a concussion may be present immediately or in the minutes or hours following injury. Symptoms may be resolved within days but can also be

Box 1

Definition of concussion in 6th International Conference on Concussion in Sport³

Sport-related concussion is a traumatic brain injury caused by a direct blow to the head, neck, or body resulting in an impulsive force being transmitted to the brain that occurs in sports and exercise-related activities. This initiates a neurotransmitter and metabolic cascade, with possible axonal injury, blood flow change, and inflammation affecting the brain.

Symptoms and signs may present immediately, or evolve over minutes or hours, and commonly resolve within days, but may be prolonged. No abnormality is seen on standard structural neuroimaging studies (computed tomography or MRI T1- and T2-weighted images), but in the research setting, abnormalities may be present on functional, blood flow, or metabolic imaging studies.

Sport-related concussion results in a range of clinical symptoms and signs that may or may not involve loss of consciousness. The clinical symptoms and signs of concussion cannot be explained solely by (but may occur concomitantly with) drug, alcohol, or medication use, other injuries (such as cervical injuries, peripheral vestibular dysfunction) or other comorbidities (such as psychological factors or coexisting medical conditions).

Data Source Patricios JS, Schneider KJ, Dvorak J, et al. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport–Amsterdam, October 2022. *Br J Sports Med* 2023;57(11):695–711. <https://doi.org/10.1136/bjsports-2023-106898>.

prolonged.³ Symptoms and signs include light and noise sensitivity, dizziness and balance disturbances, feeling tired, headache, cognitive changes in attention and concentration, foggy, mentally slowed down, and memory deficits.⁹ Other factors to consider are social and emotional disturbances including increased anxiety, posttraumatic stress disorder, chronic migraines, irritability, feeling more emotional in general, and disturbances in sleep.^{9,10}

These factors must be considered as clinicians treat concussion, begin return-to-sport protocols, and use a multimodal approach for symptom assessment.^{3,5} One widely used tool incorporating many facets of concussion recognition is the Sport Concussion Assessment Tool 6. Clinicians can use this tool to help individuals progress through the RTS progression. Early subthreshold exercise and symptom-specific rehabilitation are encouraged with a return to activities of daily living as initial priority.^{3,5,11}

Previous treatment of concussion included absolute cognitive and physical rest until asymptomatic,^{8,12–14} and is no longer recommended.^{3,15,16} Research shows the autonomic nervous system (ANS) plays a vital role in concussion recovery because it regulates critical functions such as cerebral blood flow (CBF), heart rate (HR), and blood pressure (BP),¹⁵ leading to recommendations of monitoring vitals during the recovery process.³ RTS progression has shifted from absolute rest to relative rest for the first 24 to 48 hours, to the most recent evidence encouraging light physical activity, like walking, within 24 to 48 hours of postinjury (see [Table 1](#) for updates in stepwise progression).^{3,8,9,12–14} Subsymptom aerobic exercise can help improve ANS regulation of CBF, which has been shown to help reduce symptoms including dizziness and headache in patients with persisting symptoms after concussion (also known as postconcussion syndrome).^{15,17,18} Earlier implementation of aerobic exercise can normalize physiological functions postinjury.¹⁵

A controlled stepwise return to activity has been shown to ensure that patients return to activity safely. The addition of symptom-limited activity within 24 hours and a two-step subthreshold aerobic exercise component may help athletes RTS in less time. However, individuals must ensure they work with qualified health care providers who can monitor the activity progression and not return too early or too quickly.

DISCUSSION

Return to Sport

Concussion management includes the safe return to activities of daily living (ADLs), return to learn (RTL), and RTS. RTS participation has changed since initial recommendations were made with multiple international consensus statements published from 2001, 2004, 2008, 2012, 2016, and most recently in 2022.^{3,8,9,12,13} While studies say most concussions resolve in 7 to 10 days,^{14,19} the time frame can vary widely by individual.^{14,19,20} RTS protocols have gained much attention and remain a stepwise progression of 6 steps for safe return, including light activity such as walking being introduced as early as 24 hours postinjury.³ [Table 1](#) shows the changes made regarding RTS. For the most part, Steps 5 to 6 have remained the same with most of the changes occurring in steps 1 to 2 and additional changes in the 2022 consensus statement, in steps 3 and 4.³

With RTS being individualized and sport-specific within the changing landscape of concussion management, average return-to-play timelines vary greatly. Most studies regarding RTS timelines were conducted under previous protocols.^{9,14} In 2020, a comparative analysis from the National Collegiate Athletic Association (NCAA) Concussion Study using concussion data from 1999 to 2001 (184 concussions) and

Table 1

Historical progression of return to play^{3,8,9,13,14}

Stage	Vienna 2001	Prague 2004	Zurich 2008	Zurich 2012	Berlin 2016	Amsterdam 2022
1	No activity				Symptom imited activity Relative rest for 24–48 h	Symptom limited activity Can begin within 24 h of injury
	Complete rest Recovery		Initial period of rest: 24–48 h Symptom-limited physical and cognitive rest		Gradual reintroduction of work/school—daily activities	Daily activities including <i>walking</i>
2	Light aerobic exercise					Aerobic exercise
	Stationary bike, walking—no resistance training		< 70% max predicted HR, no resistive training Increase HR		Walking or stationary bike at slow to medium pace— no resistance training	2A-LIGHT (55% max HR) 2B-MODERATE (up to 70% max HR)
3	Sport specific exercise					Individual sport specific Training away from team environment
	Skating/running drills/no head impacts Add movement					Add movement & change of direction: medical clearance prior to step 3
4	Non-contact training drills					
		Light Resistance training	More complex drills Exercise, coordination and cognitive load			Begins after resolution of symptoms Exercise to high intensity, include coordination and increased cognitive functioning
5	Full-contact practice					
	Need medical clearance		Need medical clearance Restore confidence			Medical clearance prior to step 3
6	Return to play					RTS
OTHER	If symptoms occur, wait 24 hours or until asymptomatic again and drop back to previous stage					With mild symptoms of no more than 2 points on a 0–10 point scale occurs and last for <1 h athletes may continue to progress to the next step. If symptoms are more than mild, stop and reattempt the following day.
	Vienna 2001	Prague 2004	Zurich 2008	Zurich 2012	Berlin 2016	Amsterdam 2022

From Patricios JS, Schneider KJ, Dvorak J, et al. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport—Amsterdam, October 2022. Br J Sports Med. 2023;57(11):695-711. doi:10.1136/bjsports-2023-106898.

from the NCAA-Department of Defense Concussion Assessment, Research, and Education (CARE) Consortium from 2014 to 2017 (701 concussions) showed drastic changes in concussion management and reported differences with one-third of NCAA participants reporting symptom recovery within 1 day, and 9 out of 10 players recovering within 7 days.²¹ With increased awareness of concussions including symptoms and management from 2001 to 2014, only 5% of the CARE cohort recovered within 24 hours, and nearly half reported symptoms for more than a week post injury. Studies looking at high school athletes found clearance to start light activity after periods of rest was the most significant indicator of recovery timeline.²² Stepwise protocols with individuals starting the RTS process after they were asymptomatic resulted in total days to return reported as 20.2 ± 13.9 days, with 13.6 ± 10.6 days to gain clearance for light exercise.²²

An epidemiological study looking at athletes who sustained concussion in Ivy League sports between 2013 to 2014 and 2018 to 2019 found the median time to return to limited participation was 12 days. In addition, 75% of athletes had not returned at 7 days, 42.3% had not returned in 14 days and 14.3% had not returned to limited participation at 35 days. Interestingly, the median time to return to full sport participation was 16 days, whereas 86.9% of athletes did not return within 1 week, 53% did not return within 2 weeks, and 19.4% had not returned to full participation within 35 days.²⁰

Wait and colleagues conducted a systematic review with 65 studies totaling the assessment of 21,966 patients diagnosed with concussion. RTS intervals ranged from 1 to 1820 days, and the median time to symptom resolution ranged from 2 to 11 days.²³ Overall median time to RTS post concussive injury was 21 days in 80% of published studies. **Table 2** outlines RTS intervals after concussion and various subpopulations, including pediatrics and adults, male and female, as well as level of play and concussion history. **Table 3** reports the sports that were reported and studied and RTS for mean and median for each sport. The recent advances in concussion management and recommendations of beginning light activity within 24 hours of injury show promise to decrease overall time to return to play and symptom resolution.²³

Early intervention of physical activity postinjury allows patients still experiencing symptoms to change their routine, which can help initiate healing responses. Athletes may also experience other symptoms associated with concussion, such as fear, anxiety, or depression which may worsen with the concussive injury.³ Individuals can also experience more persistent symptoms, including chronic migraines, attention problems, and sleep disturbances.¹⁰ Moreover, the longer an athlete spends not participating in their sport due to symptoms, the more deconditioned they can become. The signs and symptoms of deconditioning can mimic concussions due to alterations in CBF, which can then increase the time required to successfully return to play.^{24,25} The risk of musculoskeletal injury increases once they RTS.²⁶ Returning to some activity earlier can decrease the deconditioning effect and help resolve the common concussion symptoms and avoid prolonged recovery.^{24,25} In general, physical activity and exercise have been shown to improve sleep and various psychiatric disorders and are generally associated with better mood and quality of life.²⁷ Therefore, the early intervention and addition of physical activity prior to a full resolution of symptoms can have a positive effect on many systems of the body that effect the cascade of events that occurs with a concussive injury.

The Stepwise Progression

Stepwise progression is essential to ascertain that patients can tolerate activity within their symptoms, and adding activity too quickly can prevent the full return and lengthen the duration of the RTS process, but recently, subsymptom threshold

Table 2 Return to play ranges in days (mean and median) for subpopulations and total number of subjects outlined in the systematic review by Wait et al ²³							
	Population, n	Mean		Median		Studies	Studies
		RTP Range	Population Range	RTP Range	Population Ranges		
Articles	67						
Total subjects	21,966	6.0–149.0	15–3322	3–119	15–1715	47	40
Pediatric patients	9979	12.04–33.00	22–3322	11–38.5	27–1507	14	11
Adults	8938	7.7–34.7	15–1563	3–35	15–1715	23	20
Professional	2034	6–37.3	15–734	7–12	15–734	13	7
College	7038	6.67–34.70	23–1563	3–35	26–1715	10	10
High school	8974	11.1–28.9	54–3322	11–40	27–1507	12	11
Elementary	51	19.3	51	13	51	1	1
Male	12,098	4.8–42.3	15–2192	15–734	3–45	20	15
Female	6418	7.93–74.5	23–1130	8–119	17–950	10	10
With previous concussion	3245	10.7–40.4	6–182	14	23	3	1
Without previous concussion	7968	11.05–24.5	8–406	10	147	4	1

Abbreviation: RTP, return to play.
From Wait TJ, Eck AG, Loose T, et al. Median time to return to sports after concussion is within 21 days in 80% of published studies. *Arthroscopy* 2023;39(3):887–901. <https://doi.org/10.1016/j.arthro.2022.11.029>; **Box 1**.

Table 3
Return to play in days with reported populations and mean/median return to play ranges reported in the systematic review by Wait et al²³

Sports	Total Population Reported	Mean			Median		
		RTP Range	Population Range	Studies	RTP Range	Population Range	Studies
Football	4912	6.67–19.90	38–701	5	3–14	51–701	4
Soccer	3649	8.04–24.9	22–568	4	11–12	34–1507	3
Basketball	1794	6.85–37.3	55–463	4	7–8	55–81	2
Rugby	1527	3–22.1	24–734	5	7.8–22	35–734	4
Hockey	1369	6–17.2	15–179	4	11–18	15–329	4
Baseball/softball	666	7.9–12.7	71–207	3	9	207	1
Lacrosse	593	7.79	198	1	—	—	—
Wrestling	355	—	—	—	—	—	—
Volleyball	349	—	—	—	—	—	—
Cheerleading	163	—	—	—	—	—	—
Water polo	147	—	—	—	—	—	—
Swimming/diving	116	—	—	—	—	—	—
Australian-rules football	88	—	—	—	—	—	—
Field hockey	70	—	—	—	—	—	—
Gymnastics	55	—	—	—	—	—	—
Track/cross country	47	—	—	—	—	—	—
Tennis	17	—	—	—	—	—	—
Skiing/snowboarding	16	—	—	—	—	—	—

(continued on next page)

Table 3 (continued)							
Sports	Total Population Reported	Mean		Median		Studies	Population Range
		RTP Range	Population Range	RTP Range	Population Range		
Crew	14	—	—	—	—	—	—
Sailing	9	—	—	—	—	—	—
Fencing	5	—	—	—	—	—	—
Equestrian	4	—	—	—	—	—	—
Flag football	4	—	—	—	—	—	—
Ice skating	2	—	—	—	—	—	—
Frisbee	1	—	—	—	—	—	—
Karate	1	—	—	—	—	—	—
Mountain biking	1	—	—	—	—	—	—
Squash	1	—	—	—	—	—	—
Other	184	—	—	—	—	—	—

Abbreviation: RTP, return to play.
From Wait TJ, Eck AG, Loose T, et al. Median time to return to sports after concussion is within 21 days in 80% of published studies. Arthroscopy 2023;39(3):887–901. <https://doi.org/10.1016/j.arthro.2022.11.029>; Table 1.

aerobic exercises have been shown to help control symptoms and shorten the duration of a concussion.³ It is essential to individualize activities while considering how each patient reacts to the injury is unique. When a stimulus outweighs the body's ability to tolerate it, signs and symptoms worsen, and an increased recovery time can cause more time away from participation, causing physical and mental alterations. This can include mental health concerns, loss of identity as an athlete, and disconnect from the team. Concussion affects cognitive function and other physiological systems. Studies show that both branches of the ANS are affected within the first week after a concussive injury.²⁸ In addition, this physiological dysfunction has been said to outlast clinical measures of recovery from a concussive injury, suggesting a buffer zone of gradual increases in activity before fully returning to competition further solidifying the stepwise progression for RTS.¹⁰

The inability to regulate the ANS causes exercise intolerance, symptom exacerbation, sensitivity to carbon dioxide levels in the blood, and increased symptoms like dizziness and headaches in some athletes.¹⁵ This supports new guidelines to monitor HR and BP through recovery.³ Furthermore, the degree of early exercise intolerance and the lower HR threshold at the time of symptom exacerbation are strong indicators and predictors of time to recovery.²⁴ Understanding the role of the ANS and managing this properly during recovery with controlled and progressive aerobic exercise will improve symptoms and speed up recovery and return to play.^{15,16}

Studies have shown that subsymptom threshold exercises can help bring ANS function to homeostasis, normalizing CBF, tolerance, and symptoms.^{15,20,29} Leddy and colleagues concluded that subsymptom threshold exercise improves activity tolerance for patients with persisting symptoms after concussion and is an appropriate treatment option for their patient population, showing that starting physical activity when symptoms are present is safe.^{11,15} Assessing heart rate, blood pressure, and other markers to determine ANS function is critical, with new guidelines recommending recording systolic and diastolic blood pressure and heart rate in both supine and standing positions, with any positional changes due to the change in position noted.³

The 6th international consensus statement on concussion in sport includes adding activities of daily living and walking within 24 hours postinjury, which shows the significant evolution from a period of no activity and relative rest for 24 to 48 hours in earlier consensus statements.^{3,8,9,12-14} The stepwise return to play progression is supported in literature as a safe means to concussion recovery¹⁰ and is over 6 steps. **Table 1** outlines the newest guidelines in RTS. Each step takes at least 24 hours to complete, and the athlete may progress to the next step if symptoms do not exceed 2 points on a 0 to 10 pain scale and do not last longer than 1 hour. If symptoms exceed these limitations, individuals should stop the activity and try again the following day, understanding that each step does not equate to a day, rather each step is completed based on the individuals tolerance to activity.³ This is one of the major changes in the most recent guidelines.³ All previous RTS progressions stated that if an individual experienced symptoms at any point during their activities, they would stop, wait at least 24 hours, and resume the previous step.^{8,9,12-14}

Step 2 now includes 2 phases (2A and 2B) where health care providers challenge individuals in two ways before moving on to a sport-specific exercise. Aerobic activity is initiated at *light*, approximately 55% max HR, and then repeated at a higher intensity of roughly 70% max HR to determine readiness for sport-specific activity.³ This step in the return to play progression is supported by evidence that subthreshold exercise benefits symptom reduction and plays a major role in regulating the ANS.¹¹ Steps 1, 2A, and 2B are also considered an intervention for acute concussion.³ Step 3, the last stage in the early intervention of acute concussion and RTS, includes adding

individual sport-specific training in an environment that does not pose a risk of head impact. It is important in this step to add direction changes to ensure the athlete is ready to progress to the next step.³ Although historically, Step 3 always included sport-specific training, recent evidence adds the changes in direction.^{3,8,9,12–14}

To initiate step 4, there needs to be a complete resolution of symptoms and return to normal cognitive function both with exertion and at rest, as well as clearance from a qualified health care provider. Previously, medical clearance was necessary for step 5. The goal of step 4 is to resume the individual's normal intensity of exercise, coordination, and cognitive function and begins with noncontact drills. Progressing to step 5 includes full-contact practice. In this stage, clinicians assess the individual's readiness to participate, symptoms, cognitive function, and confidence in participation. Finally, step 6 is normal gameplay and participation in activities without limitations.³

Overall, the return to play progression from step 1 to step 6 may take at least 1 week but typically returning to sport can take up to 1 month in total.³ Although each step is said to take 24 hours typically, the amount of time spent in each step of the progression will be dependent upon the student-athlete's symptoms and the presence or absence of concussion modifiers, such as migraines, vestibular issues, attention deficit disorder (ADD/ADHD), depression, or other mental illness. The determination regarding the speed of progression through the stages may be safely augmented based on the clinical judgment of the sports medicine staff and the understanding that concussion management is very individualized. Of note, no student-athlete will be allowed back to unrestricted sports activity unless they have completed the return to learn process or can attend classes and complete schoolwork normally. Additionally, those individuals having difficulty moving on to the following stages of the RTS protocol (or RTL) steps and are not recovering into 2 to 4 weeks postinjury may benefit from additional rehabilitation strategies and a multidisciplinary approach.¹⁶

Multiple approaches to recovery depending on signs and symptoms, including cervical and vestibular rehabilitation, cognitive behavioral therapy, collaborative care, and subsymptom threshold aerobic exercise, are beneficial in concussion recovery.¹⁶ Sports medicine clinicians must understand the balance between too little and too much activity and guide their patients to a safe RTS. A systematic review highlighted various symptoms, rehabilitation modalities, and focused rehabilitation effects on time to return to play. It showed that it is essential that while athletes progress through a return-to-play process, careful consideration is given to the types of symptoms the patient is experiencing and incorporate various targeted rehabilitative therapies within the return-to-play progression.¹⁶ For instance, cervicovestibular rehab may decrease the time to RTS in athletes with dizziness, neck pain, and headaches while vestibular/oculomotor rehabilitation may reduce the time to RTS in patients with vestibular impairments such as balance deficits or convergence issues. A combination of active rehabilitation with a gradual RTS progression can facilitate a quicker and safer RTS.^{16,30}

Sport-specific considerations in exercise and the return to sport progression

Returning to activity from a concussion too soon can cause potential complications, including reinjury, physiological dysfunction, and prolonged symptoms and may predispose individuals to orthopedic injury.^{10,31} Forgoing one injury while predisposing patients to another seems counterintuitive. Some research has suggested that orthopedic rehabilitation should focus on both quantitative aspects of running load progression and qualitative characteristics of movement in competition.³² This includes highly variable, spontaneous, and unanticipated movements and reactions, deemed the conditions of *chaos* by Taberner and colleagues reflecting the unpredictable nature of sport. The controlled chaos continuum (CCC) moves from high control to high

chaos, prescribing running loads under progressively riskier conditions for injury.³² This CCC can also be considered for concussion management and recovery where each step in the stepwise progression is a way to increase the proverbial chaos or challenge of the workout. This continuum can be applied to concussion return-to-play, specifically steps 3 to 6.

Athletes with sport-related concussion show neuroanatomic and neurophysiological changes relevant to motor control even after reaching clearance, including normal neurological examination, resolution of symptoms, with no symptoms produced with RTS progression.²⁶ Altered motor function is demonstrated after sport-related concussion in muscle activation and force production, movement patterns, balance or postural stability, and motor task performance especially when paired with a cognitive task.²⁶ Based on evidence regarding the risk of musculoskeletal injuries after concussion, standard clinical assessments for athletes with a concussion should include an evaluation of neuromuscular risk factors along with physical symptoms and cognitive function before RTS.³³

Sports requiring fast reaction times will require more tailored activities like vestibular control and adaptation to activity during recovery. Neurocognitive testing can be used to help evaluate various components of performance affected by concussion, including reaction time. A consistent finding is that reaction time scores on the Immediate Postconcussion Assessment and Cognitive Testing of greater than 545 to 570 ms are associated with increased lower extremity injury risk in collegiate athletes. Chmielewski et al²⁶ discussed a conceptual framework where the increased risk of orthopedic injuries, post-concussion, come from neuroanatomical and neurophysiological changes in motor control. Examples include decreased muscle activation and force production, modified movement patterns, poor balance/postural stability, and impaired motor task performance with and without a simultaneous cognitive task.²⁶ These risk factors must be addressed during the stepwise progression through steps 3 to 5 of RTS. The progression should start with motor tasks with low physical demand, then make them more challenging and increase the cognitive and perceptual challenge mimicking the needs of the athlete's sport.²⁶ The use of neurocognitive tools can help clinicians make quantifiable decisions regarding a return to full competition, while other means of assessment for readiness to return can also include the Multiple Hop Test to examine the recovery of subtle sensorimotor impairments and related return to play readiness.³³

Another consideration is when to add resistance training to the RTS progression. This can be added to the return to play when appropriate, based on the individual and sport need. This can be considered for step 3 as lifting will increase physical demands and have the patient change body positions regularly.

There is great variability between sports in terms of cognition, speed, agility, reaction time, etc, which *control the chaos* is where sports medicine clinicians can apply their expertise of tailoring the management to the sport. This ensures tolerance of all aspects of the sport before full return to play. Step 3 of the RTS progression introduces sport-specific activities and consideration of each sport's unique requirements. Step 4 ensures a return to preinjury intensity, and step 5 focuses on gaining confidence to participate and ensure no symptoms are reproduced. Some noncontact sports may lend themselves to an earlier return by the nature of the sport such as cross country. In contrast, collision sports like football or hockey must prepare for less predictable situations such as being hit by another player. Sports medicine clinicians must consider all facets of the individual and the sport they will return to after concussion resolution when managing progression through the RTS process.

Table 4
Sports specific return for ice hockey^{3,4}

Stage	Aim(s)	Activity
1	Symptom-limited activity	Daily activities that do not provoke symptoms (eg, 10 min of slow walking)
2	Light aerobic exercises (unspecific)	a. Cardiovascular exercise on stationary bike; 25–40 min including warm up and cool down; controlled activities, low to moderate intensity (ca. 70% HR max); control for stable head/neck position while exercising b. Mobility/stretching, stabilization, and controlled balance (double and single stance) exercises
3	3.1 Aerobic exercise (progression)	a. Introduce interval training on stationary bike (or elliptical; treadmill not first choice because of head/neck impacts); 25–40 min including warm up and cool down; controlled activities, moderate intensity (ca. 70%–80% HR max); control for stable head/neck position while exercising b. Body training (no resistance/light elastic resistance) • Mobility and stretching exercises • Neck stabilization exercises (no resistance) • Trunk strength/stabilization exercises (no resistance; no explosive movements) • Basic lower/upper extremities strength exercises (light elastic resistance) in the 3 planes of movement • Balance exercises (double and single stance) on unstable surfaces No heavy resistance training.
	3.2 Ice hockey-specific exercises	a. Warm up—free skating—for 10 min at low/moderate intensity b. Basics skating with rotation/changing of direction • Skating forwards, backwards, sideways, stop & go (not explosive) • Skating in the face-off circle, figure of 8 between the two face-off circles c. Technical training with the stick/puck (1:1) • Basics: stick handling, short/long passing; easy shooting on targets No contact activities on ice. For goalkeepers: controlled and nonexplosive movements (T-glide, butterfly, half-butterfly, other); no shooting on goal
4.1	Noncontact ice hockey training drills	a. Cardiovascular training on the ice • Warm up—free skating—for 10 min at moderate intensity • Interval skating at higher intensities (ca. 90% HR max) with sufficient breaks • Cool down for 5–10 min at low intensity b. Technical training (with small group of players) • Small size game • Short/long passing

	• Shooting on goal • Selected drills with passing/shooting • Agility and reactive quickness drills c. Body training (incl. elastic resistance) • Mobility and stretching exercises • Neck stabilization exercises (elastic resistance) • Trunk strength/stabilization exercises (incl. free weights) • Basic lower/upper extremities strength exercises (elastic resistance; free weights) • Balance exercises (double and single stance) on unstable surfaces • Agility and reactive quickness drills d. Strength training • Keep resistance below about 80% 1RM, no classic weight lifting or exercises with head below the level of the hips (eg, back extensions on a bench) • Progressively increase external resistance for multi-joint exercises No contact activities on ice. For goalkeepers: controlled and explosive movements; shooting on goal 1:1 with a coach from different ranges (short to long) and angles (no slap shots)
4.2	Ice hockey training drills with controlled contact
	Participation in team training with controlled contact situations (checks on players, checks on boards) Progressively increase intensity in all training drills For goalkeepers: all movements; drills with players shooting (incl. slapshots) on goal from different angles
5	Full contact practice (team training)
	Following medical clearance, participate in normal team training a. Cardiovascular training: to be continued b. Body and strength training: regain usual routine training (unrestricted) c. Assess and assure psychological readiness
6	Return to play
	Normal match play

Abbreviations: HR max, maximum heart rate; min, minutes; RM, repetition maximum.
From Bizzini M. Optimizing performance in return to play after sport-related concussion in elite ice hockey players: a sports physical therapy and athletic trainer perspective. *Int J Sports Phys Ther* 2022;17(2):317–26. <https://doi.org/10.26603/001c.31953>; **Table 1** with permission.

Table 5
Example sport specific return strategies by sport

	Basketball	Track	Hockey	Soccer	Diving	Swim	Baseball
Step 1	Daily activity	Daily activity	Daily activity	Daily activity	Daily activity	Daily activity	Daily activity
Step 2A	15 min of bike at low intensity	Bike for everyone, typically 10–20', using HR max's	Bike 55% HR max	15–20' bike	Medium intensity walking, increased resistance biking, light swimming [no flip turns] (30-min duration), body weight strength training (15-min duration), (up to 55%	Medium intensity walking, increased resistance biking, light swimming [no flip turns, no diving into pool], (30- minute duration), body weight strength training (15- minute duration), (up to 55% of max HR)	15 min bike 55% HR max
Step 2B	40 min circuit, moderate bike with body weight sit up, push up and squats alternating	Biking for everyone, 20–30' using HR max	Bike 70% HR max	20–30' Bike	Medium intensity walking, increased resistance biking, moderate swimming [no flip turns] (30–45-min duration), light strength training (15–30-min duration), (Up to 70% of max HR)	Medium intensity walking, increased resistance biking, moderate swimming [no flip turns, no diving into pool], (30–45- min duration) light strength training (15 to 30- min duration), (up to 70% of max HR)	20 min 70% HR max

Step 3	30 min individual basketball workout with a coach	Distance/iddle distance: run—work with coach on specifics—but generally probably 30' moderate pace, Sprint/Jump: big warm up—I have an example on paper in my office, but essentially mobility drills for about 30', maybe also go for a run (ie, 1–2 miles)/lift, pole vault: warm up, pole drills (no leaving the ground or running through/ landing on the mat), maybe also go for a run/lift, throwers: footwork, light implement drills, light lift	Slide board—20' on 10 off x10/squat—rotational lunge—skater hops 20' ea for 5 minutes// On ice individual session—exertion skills (line sprints, cross ice passing—forward backward skating, 1/2 circles with shot, neutral zone dot figure 8's, neutral dots figure 8's, lie sprints	5 min bike. Super set: planks 3x1 min, Russian twists 3x20, and squats 3x12. 5L RDL airex 2x15 ea side, box jumps 3x10. T-drill 2x4. Rebound quick feet passing 2x5min. Walk-to-run 1 min walk: 4 min run for total 20–30 mins.	Return to team lift (modify plyometrics). Dryland. Diving Drills. No flips or rotations. No hand entry diving. Feet entry dives are permitted. 30–45 min in duration maximum.	Return to team lift (modify plyometrics). Dryland. 45 min–1 h swimming. Warm-up, a main-set, and warm down (all within 45 min to 1 h). No flip turns. No diving into the pool.	Lift with team, hit off tee, play catch
Step 4	NonContact practice with team	Distance/middle distance: run – higher intensity (tempo, time/ distance), Sprint/	Noncontact practice	Noncontact px: technical, bumper, no tackle pinney, shooting and	Full Team Lift. Modified practice, 1-h maximum duration. Limitation on	Full Team Lift. Modified practice, 75-min maximum duration. Warm-up, main-set, and	Full team lift, hit off machine, play catch

(continued on next page)

Table 5
(continued)

Basketball	Track	Hockey	Soccer	Diving	Swim	Baseball
	Jump: add drills (incl accel)/ approach work (as applicable), possibly short approach jumps, maybe a run as well, lift, pole vault: add run throughs (to mat), possibly 1–2 step approach, ropes/ rings, maybe a run as well, lift, Throwers: throw, starting with light implements, lift, sprint/jump: add hurdles, actual jump (as applicable), workout (depending on how long they've been out, coach may extend step 4 until ready to work out), pole vault: move through short approach to full approach, throwers: practice		finishing. No Heading.	diving difficulty level. Only dives that the athlete was proficient at prior to injury. No symptoms during or within the next 24 h after.	warm down (all within 75 min). Reintroduce flip turns.	

Step 5	Full contact practice	Full practice	Full contact practice	Full practice	Full team lift. 1–1.5-h maximum duration practice. Only dives that athlete was proficient at prior to injury. Not to add new dives in at this Step.	Full team lift. 1.5-h maximum duration practice. Warm-up, main-set, and warm down (all within 1.5 h). Reintroduce entering from the starting blocks.	Full practice
Step 6	Unrestricted activity	Unrestricted activity	Unrestricted activity	Unrestricted activity	Full Team Lift. Full practice (1.5 h and up). Can attempt dives outside of current proficiency. No symptoms will be removed from protocol	Full Team Lift. Full practice (1.5 h and up).	Unrestricted activity

Courtesy of C. Dann, MS, ATC, CSCS; P. Kula, MS, ATC; J. Doran, MS, ATC; E. Blair, MS, ATC, T.Bulford, MEd, ATC; J. Pedersen, MS, ATC; Storrs, CT.

Considering this, Bizzini and colleagues proposed a step-by-step progression for ice hockey that recommends 4 to 5 full team training practices, including unrestricted contact, before being considered mentally and physically ready for participation in competition.³⁴ **Table 4** outlines the return to play and stepwise progression for ice hockey players.³⁴

Table 5 gives examples of sport-specific progression options for clinicians developed by a group of athletic trainers at a Division I University based on their expertise. These options incorporate lifting, sport-like activity, and return to practice.

Special Considerations

In 2019, the NCAA settled a lawsuit about concussion monitoring in all NCAA sports.³⁵ It was agreed that the NCAA would create a reporting process of concussion diagnosis and time to resolution, as well as provide resources for screening and medical evaluations associated with sports-related concussions. This included required annual educational materials for all coaches, student-athletes, administrators, and medical staff members. The settlement requires all schools to conduct pre-season baseline screenings before any competition, no athlete with a concussion diagnosis can RTS on the same day and must be cleared for participation by a physician. Additionally, medical personnel with training in concussion diagnosis, treatment, and management must be present at all contact sports competitions and available for all contact sports practices. The terms of this settlement laid the groundwork for participating schools to have guidelines for concussion protocols that they may not have had prior. This is meant to aid in the early recognition and treatment of concussions moving forward. As evidence continues to evolve, health care providers must ensure they stay current with the most recent evidence.

SUMMARY

Concussion research is evolving and staying up to date with the most recent evidence on safe management is essential. One thing remains constant: early recognition of concussive injuries and early intervention is necessary to ensure that individuals recover and return to preinjury status before returning to competition. The emphasis on sport-related concussion has brought on many positive changes and increased awareness of concussion. In this way, more athletes are seeking medical attention, and physicians are more involved in management.⁷

Individualization of the treatment is key in tailoring the return process to fully address symptoms and sport-specific needs. Frequent monitoring of symptoms and early, controlled exercise can help student athletes recover from their injury and RTS faster than previously understood. Providing an optimal RTS after concussion contributes to the health and overall well-being of the athlete.

CLINICS CARE POINTS

- Early recognition and access to sports medicine clinicians is essential in concussion management.
- Research has shown that subsymptom threshold exercises benefit the athlete in terms of recovery.
- Incorporating sport-specific movements into the RTS process helps mitigate the risk of injury upon returning from concussion.

DISCLOSURES

The authors have nothing to disclose.

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